

ARSUM NADEEM CHAUDHARY

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EDUCATION

Pomona College | Claremont, CA (*Full Financial Aid Scholar*)

May 2025

Bachelors of Arts in Computer Science and Mathematics (Double Major)

GPA: 3.8

Coursework: Data Structures, Algorithms, Big Data, Complex Systems, AI, Neural Nets, Operations Research, Adv Linear Algebra

SKILLS

Languages: Python, JavaScript, Java, SQL

Frontend & Backend: React, Node.js, Flask, REST APIs, Data Modeling

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, S3), Google Cloud Functions, Vercel

Data & AI: Pandas, TensorFlow, Snowflake, LLM Application Development (LangChain), Workflow Orchestration (n8n)

EXPERIENCE

Software Engineer (Full Stack) - [PyqAI \(YC W23\)](#)

June 2025 - Jan 2026

- Designed and built core **self-serve backend** and cloud infrastructure for [Mulligan](#), including third-party API integrations and usage-based billing systems (Stripe + Metronome) supporting **production SaaS workflows**
- Automated **~90%** of insurance back-office workflows via Dockerized RPA services, enabling **200+** brokers to quote policies **6x faster** (<10 min vs 1 hr) and saving clients an estimated **~\$700K** annually
- Identified demo latency as the primary bottleneck through user testing, then **reduced end-to-end latency 10x (150s → 15s)** by parallelizing distributed cloud functions across AWS and Google Cloud
- Designed **stateful multi-agentic workflow orchestration** to integrate unreliable insurance carrier APIs, handling schema normalization, retries, partial failures, and deterministic end-state guarantees at scale

Software Engineering Intern - [Learning Economy Foundation](#)

Sep 2023 - Dec 2023

- Streamlined credential generation for **1k+ daily users** by building automated backend preview pipelines using OpenAI function calling, increasing credential issuance by **200%**, improving learner retention by **15%**, and reducing engineering overhead by **10+ hours/week**

Data Engineering Intern - [Inabia Software & Consulting](#)

June 2023 - Aug 2023

- Enabled pharmaceutical clients (e.g., Allergan Aesthetics) to ensure data integrity for compliance audits and boost processing efficiency by **20%** by building ETL pipelines to ingest/normalize **1TB** of sensitive health data
- Improved client-facing communication and **regulator trust** by working with compliance teams to develop NLP models to flag problematic advertising language (e.g., unverified claims)

LEADERSHIP

Lead College Application Coach | Pomona College Academy for Youth Success ([PAYS](#))

May 2022 - Aug 2024

- Led team of **15** to coach **92** underrepresented high-school students over **3 summers** in Pomona's intensive three-year equity mentorship program for STEM & college applications, contributing to placements at **top US universities** and **\$760k** secured in financial aid and scholarships
- Taught liberal arts-style critical thinking, communication, and problem-solving skills by creating a first-time **Math and Music module**, preparing students to thrive in selective university environments beyond technical coursework

RELEVANT PROJECTS

Brain Tumor Classification Model - [GitHub](#)

Jan 2025 - Feb 2025

- Enabled real-time tumor classification (**<200ms per image, ~96% accuracy**) by designing a high-throughput pipeline with an Xception + custom CNN ensemble to preprocess 1k+ MRI scans
- Built **saliency-maps** to visualize predictions, improving **clinician trust** in AI medical decision-making
- Presented the applied AI solution to **Google** and **healthcare leaders** at an accelerator program

Twitter Clone Web Application - [GitHub](#)

May 2024 - June 2024

- Built a **full-stack social platform** supporting **1M+** posts by designing optimized database schemas and indexed queries.
- Implemented **REST APIs** for posts, likes, follows, and authentication to support high-throughput reads and writes.
- Deployed scalable services using Docker, Nginx, and Gunicorn on AWS behind a **load balancer**

Coronavirus Twitter Analysis Project - [GitHub](#)

Jan 2024 - Feb 2024

- Analyzed **1.1B tweets** in 10+ languages across 20+ countries to create **4 critical visuals**, highlighting **cultural and linguistic differences** in user reactions and demonstrating the power of **data storytelling** for empathy at global scale

ADDITIONAL INFORMATION

Involvement: Team Lead: UCLA DataFest, 4x AI Hackathons; College Speech & Debate; Music (Balinese Gamelan, solo flutist)

Honors: Silicon Valley Program Scholar; Pomona College Scholar (multiple semesters, 4.0 GPA); Full Financial Aid International Scholar; Sigma Xi Research Honor Society (Math Thesis); Founded First Library at Human Rights Activist Mukhtar Mai's School